

Prefab:-

1. Compare the two single-ended differential amplifiers discussed in this lab. Rate the differential gain, common-mode gain, output voltage swing range, CMRR and common-mode input voltage range. Include expressions for each design specifications.

SIMPLE Differential Amplifier.

Metric.

Cascode current mirror load + cascode tail.

1. Differential gain : $A_{DM} = g_{m1}(r_{o2} || r_{oc})$
 A_{DM} . (active load sets r_{out})

$A_{DM} \approx g_{m1} r_{out}$ with r_{out} much larger because cascodes increase output resistance.

2. Common-mode gain : $A_{CM} = - \frac{1}{2r_{o,tail} \cdot g_{m,load}}$

Similar formula but tail r_o much larger with cascode tail $\rightarrow$ much smaller A_{CM}

3. CMRR : $20 \log_{10} \left(\frac{A_{DM}}{A_{CM}} \right) \rightarrow$ modest for simple mirror.

Much larger CMRR because cascodes raise r_{out} and raise tail r_o .

4) Output voltage : Larger for simple mirror, (fewer stacked device)

Reduced (stacking of cascode devices reduces headroom.

Bandwidth / GBW : g_{m1} / C_L (dominant pole at output) - needs g_m to meet GBW)

Same small signal expression (dominant pole @ high Σ -node) but larger gain may reduce full-power bandwidth; g_m requirement for GBW unchanged

slew rate

$$SR = \frac{I_{tail}}{C_L}$$

$$SR = \frac{I_{tail}}{C_L}$$

2) Design a simple differential amplifiers to obtain the following specification,

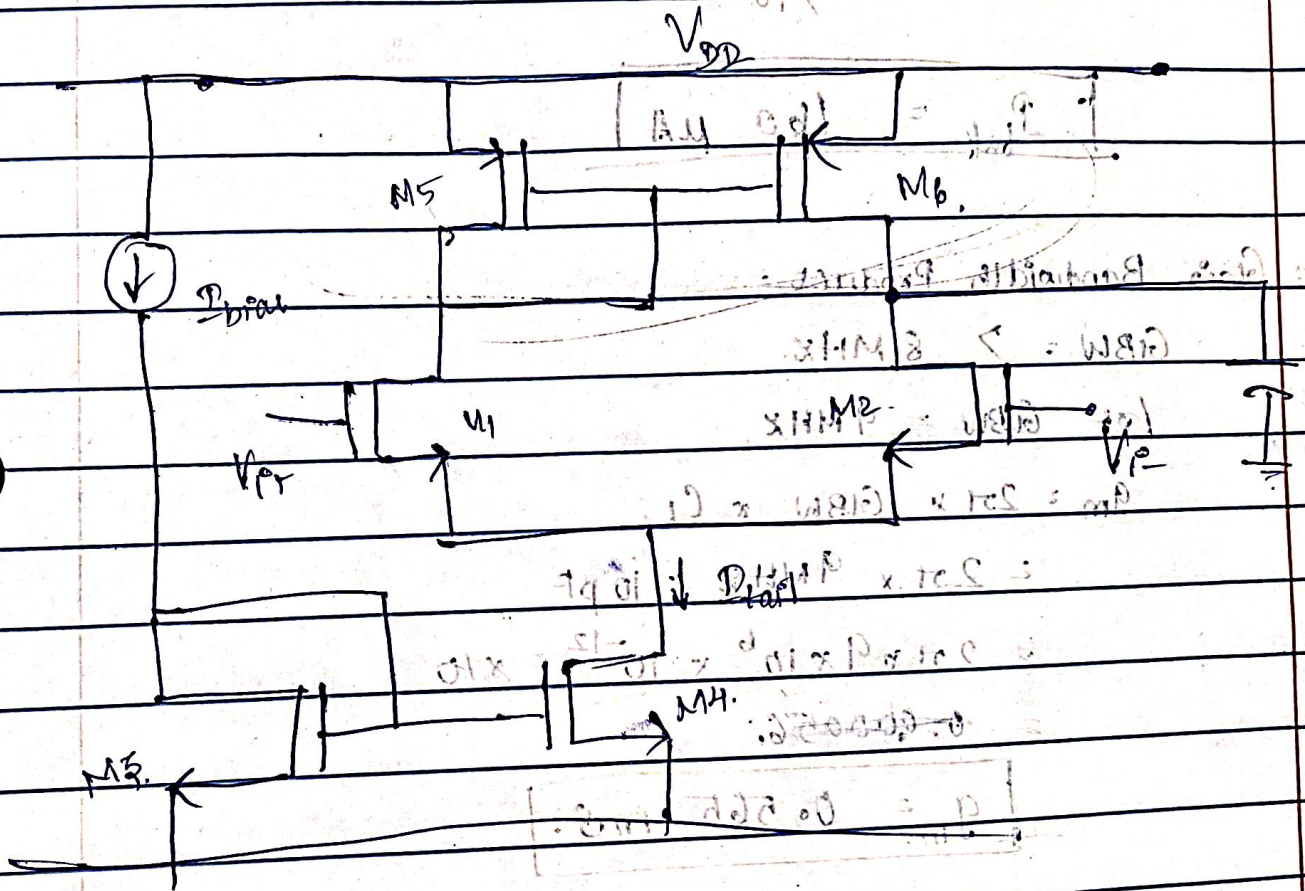
Slew Rate $> 15 \text{ V}/\mu\text{s}$

Gain-Bandwidth Product $> 8 \text{ MHz}$

Common-mode Input voltage range $> 0.4 \text{ V}$

Power supply $V_{DD} = 1.8 \text{ V}$, $V_{SS} = -1.8 \text{ V}$

Load Capacitance $= 10 \text{ pF}$



Starting with the slew rate requirement:

Step 1:-

$$> 15 \text{ V}/\mu\text{s}$$

$$C_L = 10 \text{ pF}$$

Assume $SR = 16 \text{ V}/\mu\text{s}$

$$SR = \frac{P_{tail}}{C_L}$$

$$\frac{P_{tail}}{C_L} = 16 \text{ V}/\mu\text{s}$$

$$P_{tail} = 16 \text{ V} / 10^{-6} \times 10 \times 10^{-12}$$

$$P_{tail} = 160 \mu\text{A}$$

Step 2: Gain Bandwidth Product:-

$$GBW > 8 \text{ MHz}$$

Let $GBW = 9 \text{ MHz}$

$$g_m = 2\pi \times GBW \times C_L$$

$$= 2\pi \times 9 \text{ MHz} \times 10 \text{ pF}$$

$$= 2\pi \times 9 \times 10^6 \times 10^{-12} \text{ F} \times 10$$

$$= 0.00056$$

$$g_m = 0.565 \text{ mS}$$

Transconductance and Drain Current Relationship

$$g_m = \sqrt{2k_N I_D}$$

$$k_N = \frac{g_m^2}{2I_D}$$

$$I_D = \frac{I_{Dsat}}{2} = \frac{160 \mu A}{2} = 80 \mu A$$

$$k_N = (0.565)^2$$

$$(0.565 \times 80 \mu A)^2$$

$$= \frac{(0.565 \times 10^{-3})^2}{2 \times 80 \mu A}$$

$$= 1.995 \times 10^{-3} A/V^2$$

$$k_N = \mu_n C_{ox} \frac{W}{L}$$

$$214.326 \mu A/V^2$$

$$1.995 \times 10^{-3} = 214.326 \times 10^{-6} A/V^2 \times \frac{W}{L}$$

$$\frac{1.995 \times 10^{-3}}{214.326 \times 10^{-6}} = \frac{W}{L}$$

$$9.30$$

$$\left(\frac{W}{L} \right) = 9.30$$

$$\left(\frac{W}{L} \right) \approx 10$$

for M_1, M_2

5, Common - Mode Input Voltage Range,

$$V_{SS} + \sqrt{\frac{2P_{tail}}{k_N(w/L)_{3,4}}} + \sqrt{\frac{P_{tail}}{k_N(w/L)_{1,2}}} + V_{TN} < V_{in(CM)} <$$

$$V_{TN} = 0.4470368 \text{ V}$$

$$\sqrt{\frac{P_{tail}}{k_p(w/L)_{5,6}}} = |V_{TP}| + V_{TN}$$

$$V_{TP} = 0.4216376 \text{ V}$$

$$k_p = \mu_n C_{ox} = (\text{from lab})$$

$$k_p = 47.52 \mu\text{A/V}^2$$

Lets assume,

$$(w/L)_{3,4} = 22$$

$$(w/L)_{5,6} = 29$$

$$V_{in, min} = -0.9 + \sqrt{\frac{2 \times 160 \mu\text{A}}{214.326 \left(\frac{w}{L}\right)_{3,4}}} + \sqrt{\frac{1.2 \times 160 \mu\text{A}}{47.52 \left(\frac{w}{L}\right)_{5,6}}} + V_{TN}$$

$$= -0.9 + \sqrt{0.06786} + \sqrt{0.05148} + 0.447$$

$$= 2.0807 + 0.447$$

$$V_{in, min} = 0.03439$$

$$V_{in(min)} = 2.0807 + 0.447$$

$$= 2.5277$$

$$V_{max} = V_{DD} - \sqrt{\frac{P_{tail}}{k_p \left(\frac{W}{L}\right)_{56}}} - |V_{fp}| + V_{TN}$$

$$= 0.9 - \sqrt{\frac{160 \mu A}{47.52 \times 29.0}} - |0.421| + 0.447$$

$$= 0.9 - 0.3407 + 0.026$$

$$V_{max} = 0.5853$$

$$V_{max} - V_{min} = 0.5853 - 0.03439$$

$$0.55091 > 0.4V$$

The parameters:

$$\left(\frac{W}{L}\right)_{3,4}^{18} = 22$$

$$A_{1,4} = \left(\frac{W}{L}\right)_{56}^{29} = 29$$

$$P_{tail} = 160 \mu A$$

3. Design the differential amplifier with cascode current mirrors. In figure 6.18 to obtain the following specification.

$$\text{Slew Rate} > 7 \text{ V}/\mu\text{s}.$$

$$\text{Gain Bandwidth Product} > 8 \text{ MHz}.$$

$$\text{CMRR} > 65 \text{ dB}.$$

$$\text{Power Supply } V_{DD} = +V_{SS} = 0.9 \text{ V}.$$

$$\text{Load Capacitance} = 10 \text{ pF}.$$

Step 1: Slew-rate $\rightarrow$ choose tail current, I_{tail}

$$SR = \frac{dV_{out}}{dt} = \frac{I_{tail}}{C_L}$$

$$I_{tail} = SR \times C_L$$

$$\text{Let Slew rate} = 8 \text{ V}/\mu\text{s}.$$

$$I_{tail} = 8 \text{ V}/\mu\text{s} \times 10 \times 10^{-12} \text{ F}$$

$$= 8 \times 10^{-5} \text{ A}$$

$$= 80 \mu\text{A}$$

Choosing the margin. Assume $I_D = 40 \mu\text{A}$

$$I_{tail} = 90 \mu\text{A}$$

$$SR = 90 \mu\text{A} / 10 \text{ pF} = 9 \text{ V}/\mu\text{s} \text{ meets spec}$$

Step 2

GBW $\rightarrow$ required Input gm,

$$GBW = 8 \text{ MHz}$$

$$g_{m, \text{req}} = 2\pi \times GBW \times C_L$$

$$= 2\pi \times 8 \times 10^6 \times 10 \times 10^{-12}$$

$$= 502.65 \mu\text{S}$$

Let $I_D = 40 \mu\text{A}$

$$V_{ov} = \frac{2 I_D}{g_m} = \frac{2 \times 40 \mu\text{A}}{502.65 \mu\text{S}}$$

$$V_{ov} = 0.159156 \text{ V}$$

W/L for M_1, M_2 $K_n = 214.326$

$$g_m \rightarrow \frac{W}{L} = \frac{g_m^2}{2 K_n I_D}$$

$$= \frac{(502.65 \times 10^{-6})^2}{2 \times 214.326 \times 10^{-6} \times 40 \times 10^{-6}}$$

$$= 14.7355$$

$$\left(\frac{W}{L} \right)_{M_1, M_2} \approx 15$$

$$W/L \approx 15$$

Step 3.

$$CMRR \geq 65 \text{ dB.}$$

$$A_{CM} \approx 1$$

$$2 \cdot r_o, \text{ tail } g_m, \text{ load}$$

$$g_{m, \text{ load}} = 250 \mu\text{S.}$$

$$r_{o, \text{ tail}} > 1$$

$$2 \cdot g_{m, \text{ load}} = |A_{CM, \text{ max}}|$$

$$r_{o, \text{ tail}} = \frac{1}{g_{m, \text{ load}}} = 22.4 \text{ k}\Omega$$

$$2 \times 250 \times 10^{-6} \times 0.089$$

Assuming, M_3, M_4 , for fixed $L = 270 \text{ nm}$

$$\frac{W}{L} \approx 8$$

$$M_5, M_6 \Rightarrow \frac{W}{L} \approx 12$$

$$M_7, M_8 \Rightarrow \frac{W}{L} \approx 12$$

$$M_9 \Rightarrow \frac{W}{L} \approx 10$$

$$M_{10} \Rightarrow \frac{W}{L} \approx 12$$